

DAHYE KIM

Department of Computer Science, Boston University, USA
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RESEARCH INTERESTS

Image and Video Synthesis; Diffusion Models; Multimodal Models; Video Understanding; Interpretability

EDUCATION

Boston University <i>PhD Student in Computer Science</i> • Advisor: Prof. Deepti Ghadiyaram	MA, United States <i>Sep. 2024 - Current</i>
Yonsei University <i>M.S. of Electrical & Electronic Engineering</i> • Advisor: Prof. Kwanghoon Sohn	Seoul, South Korea <i>Sep. 2022 - Aug. 2024</i>
Yonsei University <i>B.S. of Electrical & Electronic Engineering</i> • Cumulative GPA: 4.0/4.0 (Class Rank: 2/224)	Seoul, South Korea <i>Mar. 2020 - Aug. 2022</i>
Hongik University <i>Electronic & Electrical Engineering</i>	Seoul, South Korea <i>Mar. 2018 - Feb. 2020</i>

PREPRINTS

1. **Dahye Kim**, Bhuvan Sachdeva, Karan Uppal, Naman Gupta, Vineeth N. Balasubramanian, and Deepti Ghadiyaram, “Swift Sampling: Selecting Temporal Surprises via Taylor Series,” *arXiv preprint arXiv:2605.22678*, 2026.
2. **Dahye Kim** and Deepti Ghadiyaram, “Concept Steerers: Leveraging K-Sparse Autoencoders for Test-Time Controllable Generations,” *arXiv preprint arXiv:2501.19066*, 2025.

PUBLICATIONS

1. **Dahye Kim**, Deepti Ghadiyaram, and Raghudeep Gadde, “DDiT: Dynamic Patch Scheduling for Efficient Diffusion Transformers,” In *CVPR*, 2026 (Highlight).
2. **Dahye Kim***, Xavier Thomas*, and Deepti Ghadiyaram, “*Revelio*: Interpreting and leveraging visual semantic information in diffusion models,” In *ICCV*, 2025; *CVPRW MIV*, 2025 (Oral).
3. **Dahye Kim**, Jungin Park, Jiyoung Lee, Seongheon Park, and Kwanghoon Sohn, “Language-free Training for Zero-shot Video Grounding,” In *WACV*, 2023 (Oral).
4. Seongheon Park, Hanjae Kim, Minsu Kim, **Dahye Kim**, and Kwanghoon Sohn, “Normality Guided Multiple Instance Learning for Weakly Supervised Video Anomaly Detection,” In *WACV*, 2023 (Oral).

RESEARCH EXPERIENCES

Boston University <i>Research Assistant (Advisor: Prof. Deepti Ghadiyaram)</i> • Generative Models; Multimodal Models; Interpretability	MA, United States <i>Sep. 2024 - Current</i>
Amazon <i>Applied Scientist Intern (GenAI team, Mentor: Raghudeep Gadde)</i> • Efficient Diffusion Models	WA, United States <i>Jun. 2025 - Nov. 2025</i>
Digital Image Media Lab., Yonsei University <i>Research Assistant (Advisor: Prof. Kwanghoon Sohn)</i> • Multi-modal Learning with Visual and Textual Representation	Seoul, South Korea <i>Sep. 2022 - Jul. 2024</i>

Digital Image Media Lab., Yonsei University

Undergraduate Research Assistant (Advisor: Prof. Kwanghoon Sohn)

- Human-object Interaction (HOI) Detection; Video Understanding

Seoul, South Korea

Jan. 2021 - Aug. 2022

AWARDS AND HONORS

Honors

- High Honors Student (Top 3% of department) *Fall 2020, Spring 2021, Fall 2021*
- Honors Student (Top 10% of department) *Spring 2020*

Scholarships

- Brain Korea 21 (BK21) Scholarship, National Research Foundation of Korea *Sep. 2022 - July. 2024*
- Automotive System IC Scholarship,
Automotive System IC Fusion Human Resource Research Center *Sep. 2022 - July. 2024*
- Teaching Assistant Scholarship, Yonsei University (\$ 3,500 total) *Fall 2022, Fall 2023*
- Graduate Student Research Assistant Scholarship, Yonsei University (\$ 3,500) *Fall 2022*
- Academic Excellence Scholarship, Yonsei University (\$ 3,600 total) *Spring 2021, Fall 2021*
- National Science and Technology Scholarship (\$ 3,600),
National Research Foundation of Korea *Fall 2020*
- Academic Excellence Scholarship, Hongik University (\$ 6,300 total) *Fall 2018, Spring 2019, Fall 2019*

PROJECTS

Yonsei University-Yonsei Signature Research Cluster

Sep. 2022 - Jul. 2024

- Project Title: Development of Multimodal-based General-purpose Social Artificial Intelligence Technology
- Scope: Multi-Modal Learning, Zero-Shot Learning, Meta-Learning

Ministry of Science and ICT, Mid-Level Research

Sep. 2022 - Aug. 2023

- Project Title: Development of Complex Situational Awareness and Prediction Technology through Multi-Modal Data Fusion and Social Artificial Intelligence
- Scope: Multi-modal Learning, Video Understanding

Korea Institute of Science and Technology (KIST)

Sep. 2022 - Aug. 2023

- Project Title: Deep Identification and Tracking of Missing Person in Heterogeneous CCTV
- Scope: Object Detection, Object Tracking, Person Re-Identification, Anomaly Detection

SERVICES

Workshop Organizer

- XAI4CV: Explainable Artificial Intelligence for Computer Vision (CVPR 2025, CVPR 2026)

Reviewer

- Mechanistic Interpretability for Vision (MIV) at CVPR 2025
- AI for Content Creation (AI4CC) at CVPR 2026
- IJCV 2025
- CVPR 2026
- ECCV 2026

TEACHING EXPERIENCES

CS 541: Applied Machine Learning

Spring 2025

EEE 3313: Introductory Digital Labs

Fall 2023

EEE 2060: Signals and Systems

Fall 2022

SELECTED COURSEWORK

Artificial Intelligence: Introduction Artificial Intelligence (A), Intelligent Control (A), Multimodal Deep Learning (A), Digital Image Processing (A), Medical Imaging System and its Application for Artificial Intelligence (A), Topics in Computer Vision (A), and Special Topics for Deep Learning (A)

Mathematics: Linear Algebra (A), Mathematical Statistics (A), Probability and Random Variables (A), Random Process (A), and Information Theory (A)

Programming: Data Structure and Algorithms (A), and Operating Systems (A)

OTHER SKILLS

Language and Tools: Python, MATLAB, C/C++, PyTorch, Linux Shell, LaTeX

Language Ability: English (fluent), Korean (native)